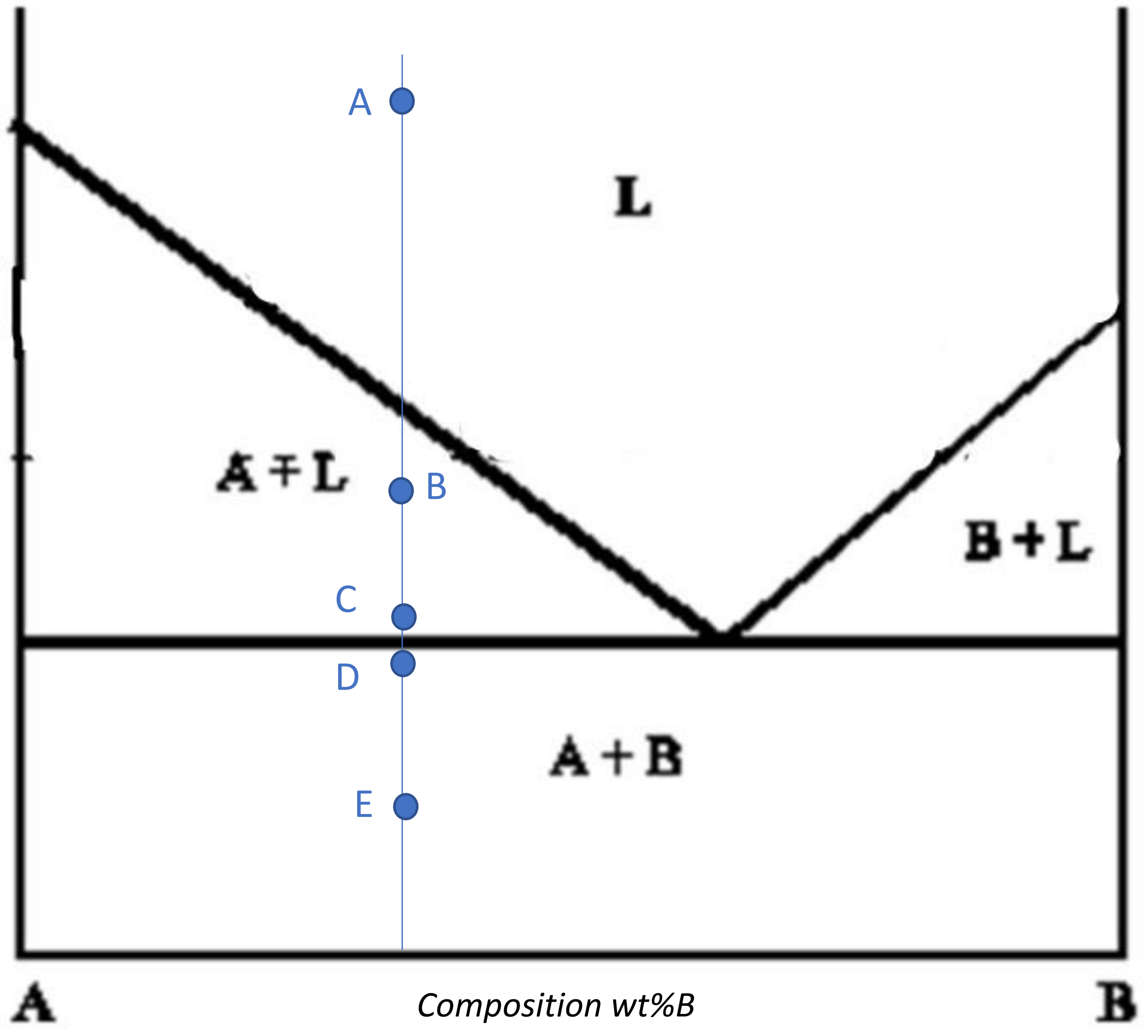
STM Exercise 6

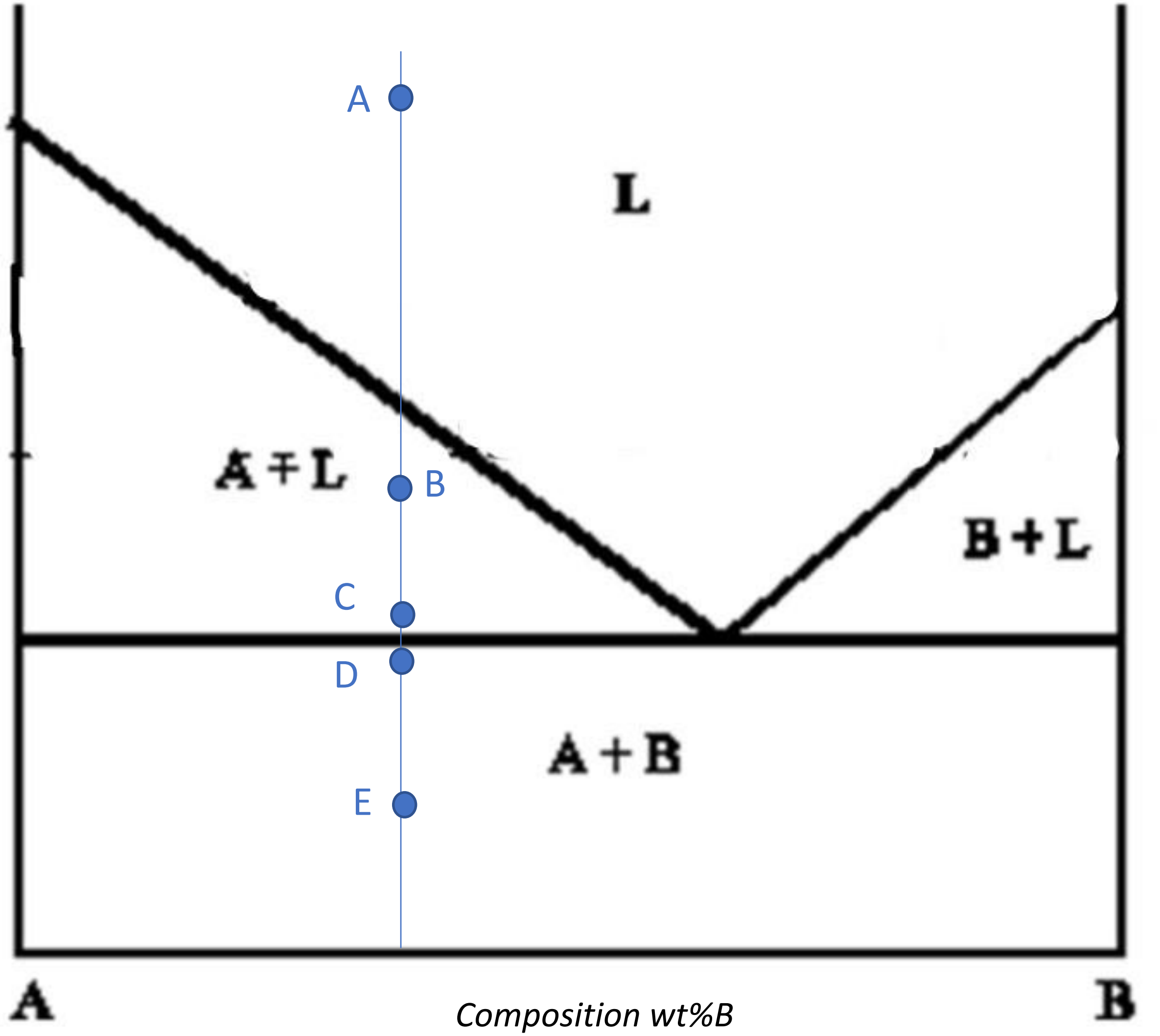
Phase Diagrams

(no solubility in the solid state)

(partial solubility in the solid state)

Exercise 1: determine the phase present, their composition and relative amounts in the point A, B, C (at the beginning of the eutectic transformation), D (at the end of the eutectic transformation) and E indicated in the phase diagram. Schematize the microstructure in each point.



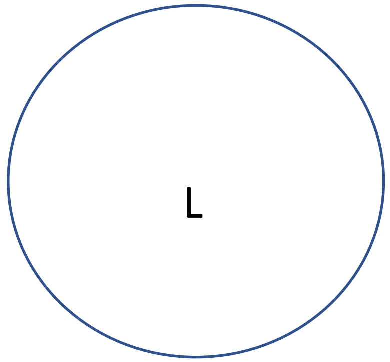


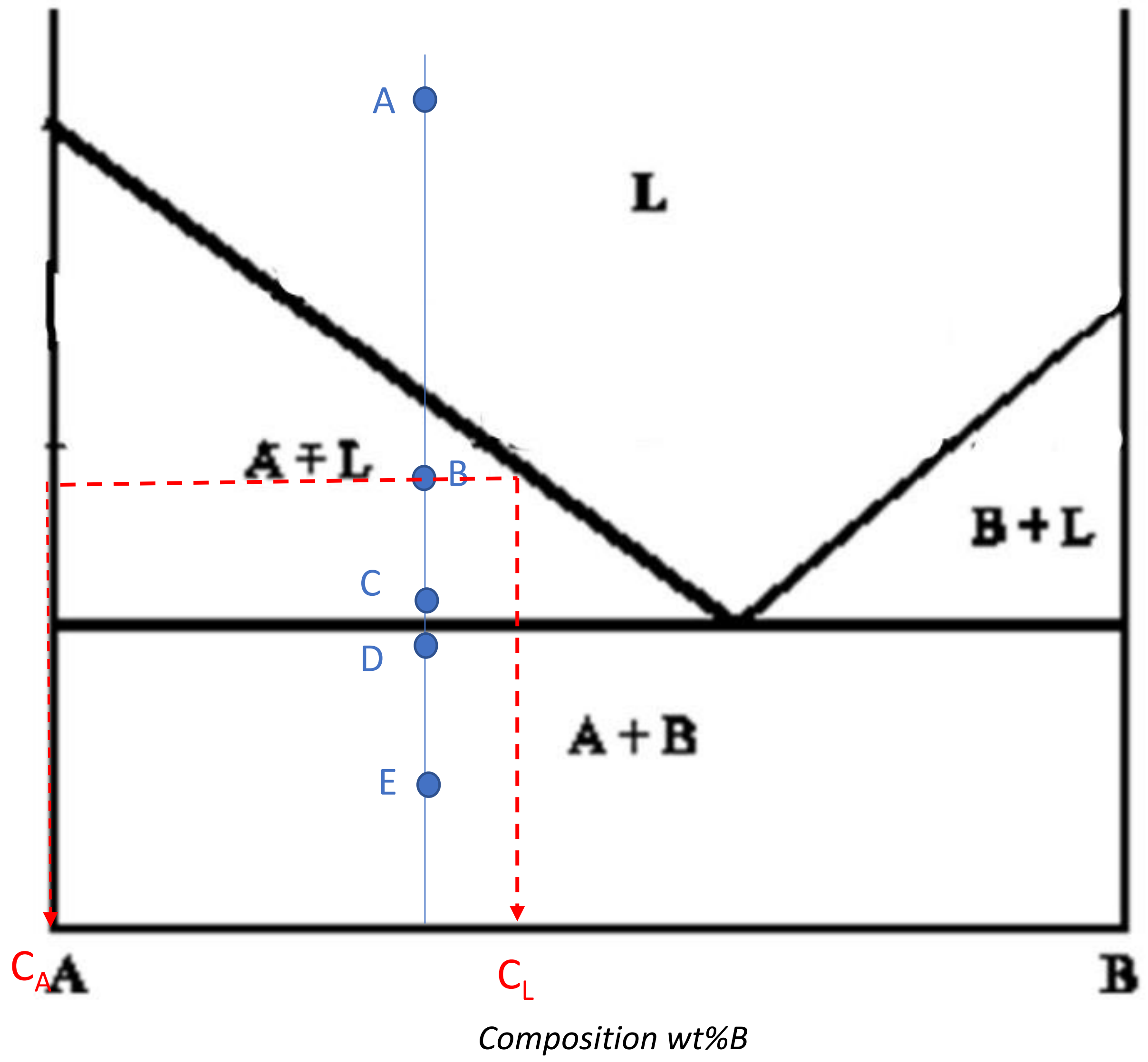
Point A:

Phases: 1 phase liquid

Composition: 35%B, 65%A

Relative amount: 100% liquid phase





<u>Point B:</u>

Phases: 2 pahses solid (A) +liquid (L)

Composition:

2 phases region → <u>horizontal rule</u>

Composition of liquid (L): C_L= 45% B, 55%A

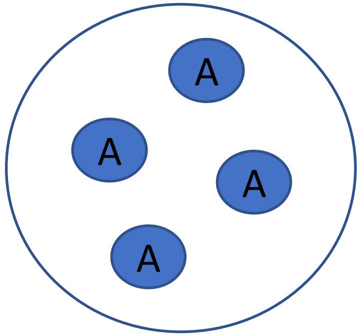
Composition of solid (A): C_A= 100%A

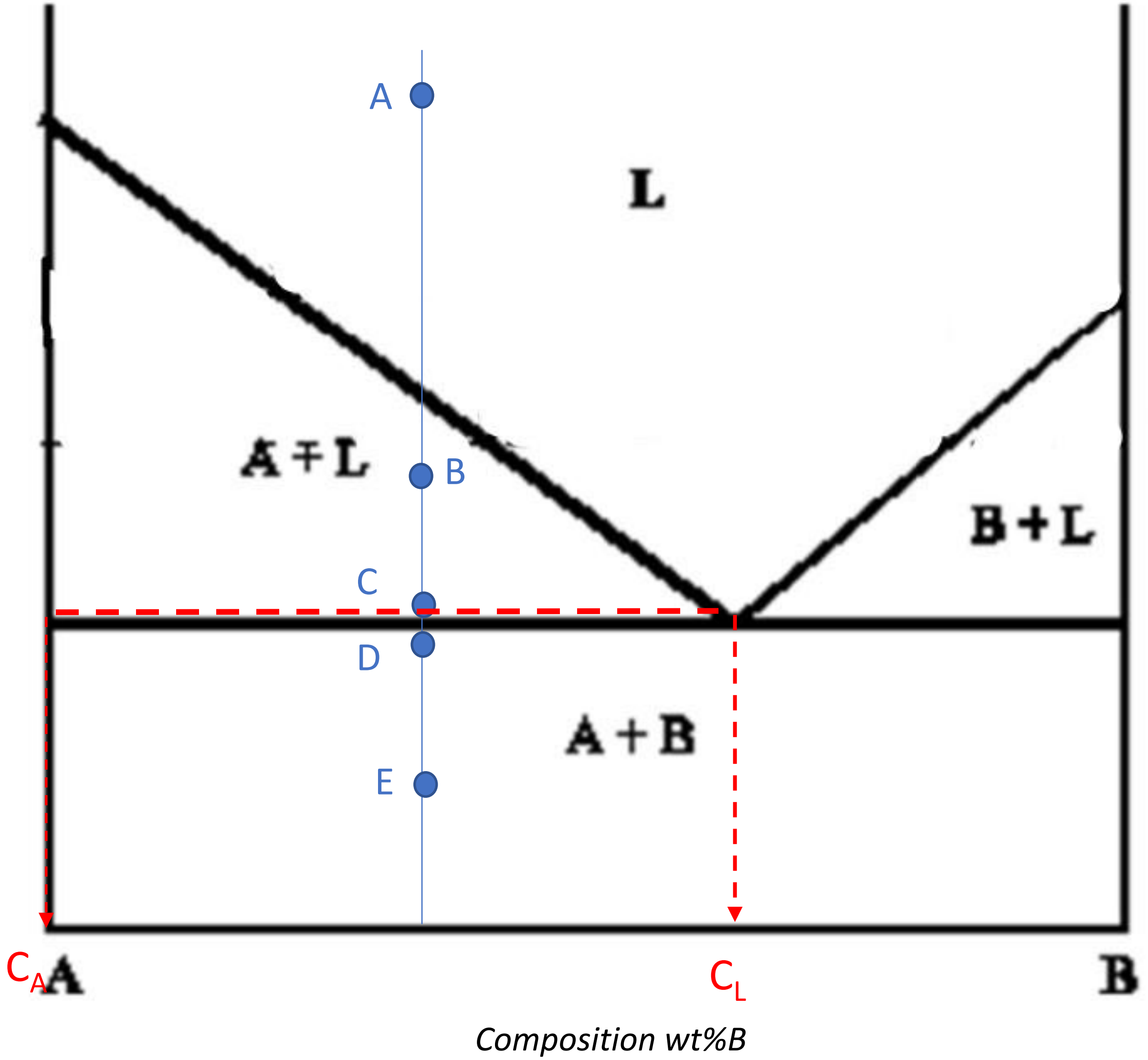
Relative amount:

2 phases region → <u>lever rule</u>

% A= (45-35)/(45-0)*100=22%

%L=(35-0)/(45-0)*100=78%





Point C:

Phases: 2 pahses solid (A) +liquid (L)

Composition:

2 phases region → horizontal rule

Composition of liquid (L): C_L= 60% B, 40%A

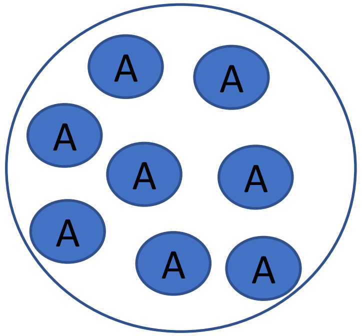
Composition of solid (A): C_A= 100%A

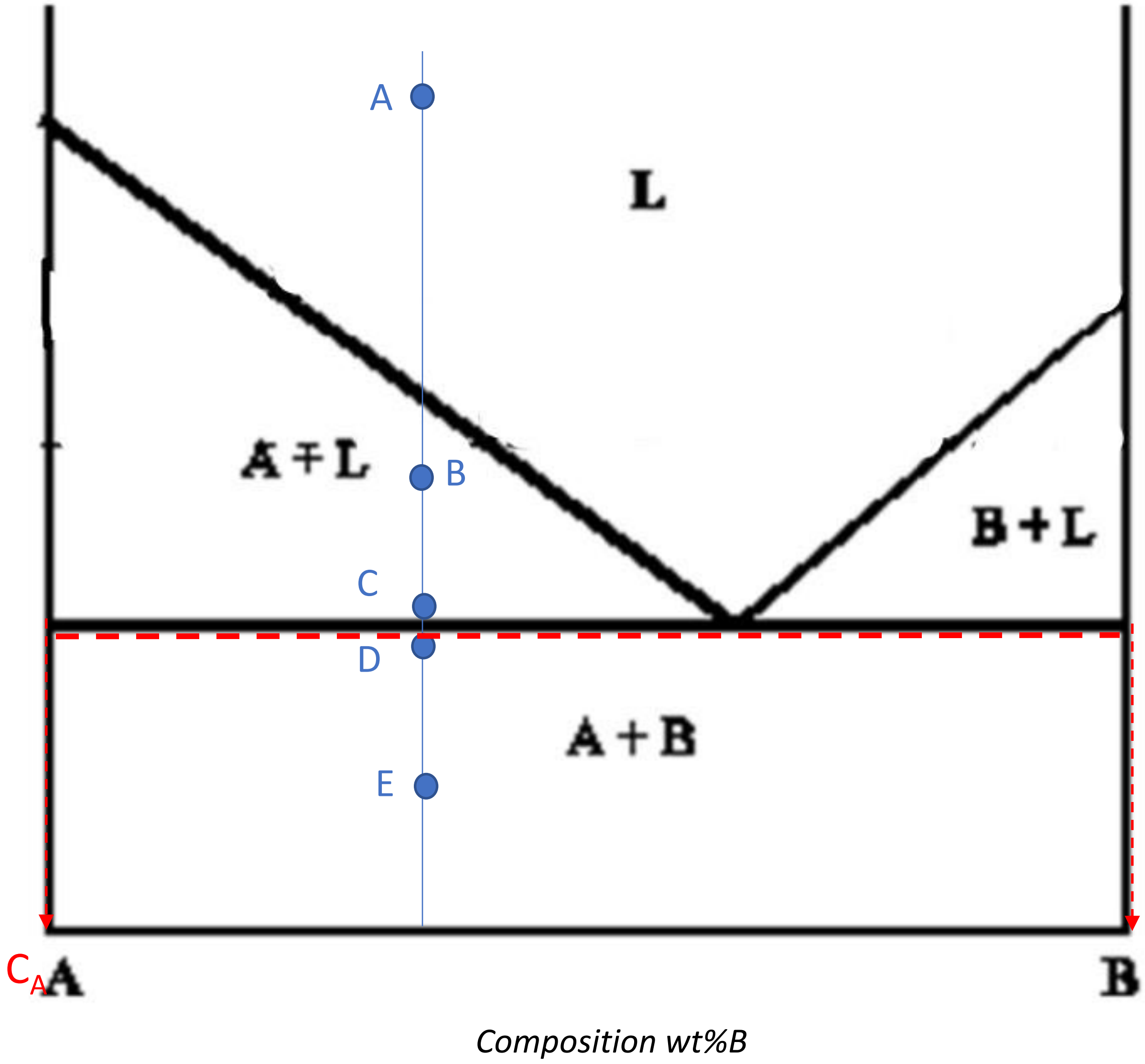
Relative amount:

2 phases region → lever rule

% A = (60-35)/(60-0)*100 = 42%

%L = (35-0)/(60-0)*100 = 58%





Point D:

Phases: 2 pahses solid (A) + solid (B)

Composition:

2 phases region → horizontal rule

Composition of Solid (A): C_A= 100%A

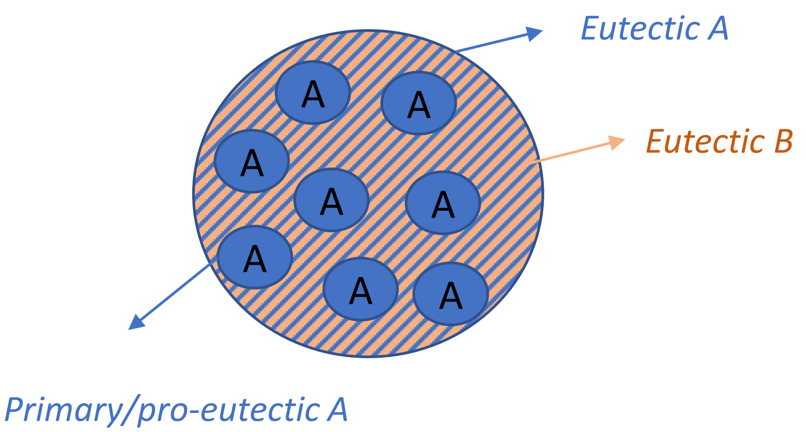
Composition of solid (B): C_B= 100%B

Relative amount:

2 phases region → lever rule

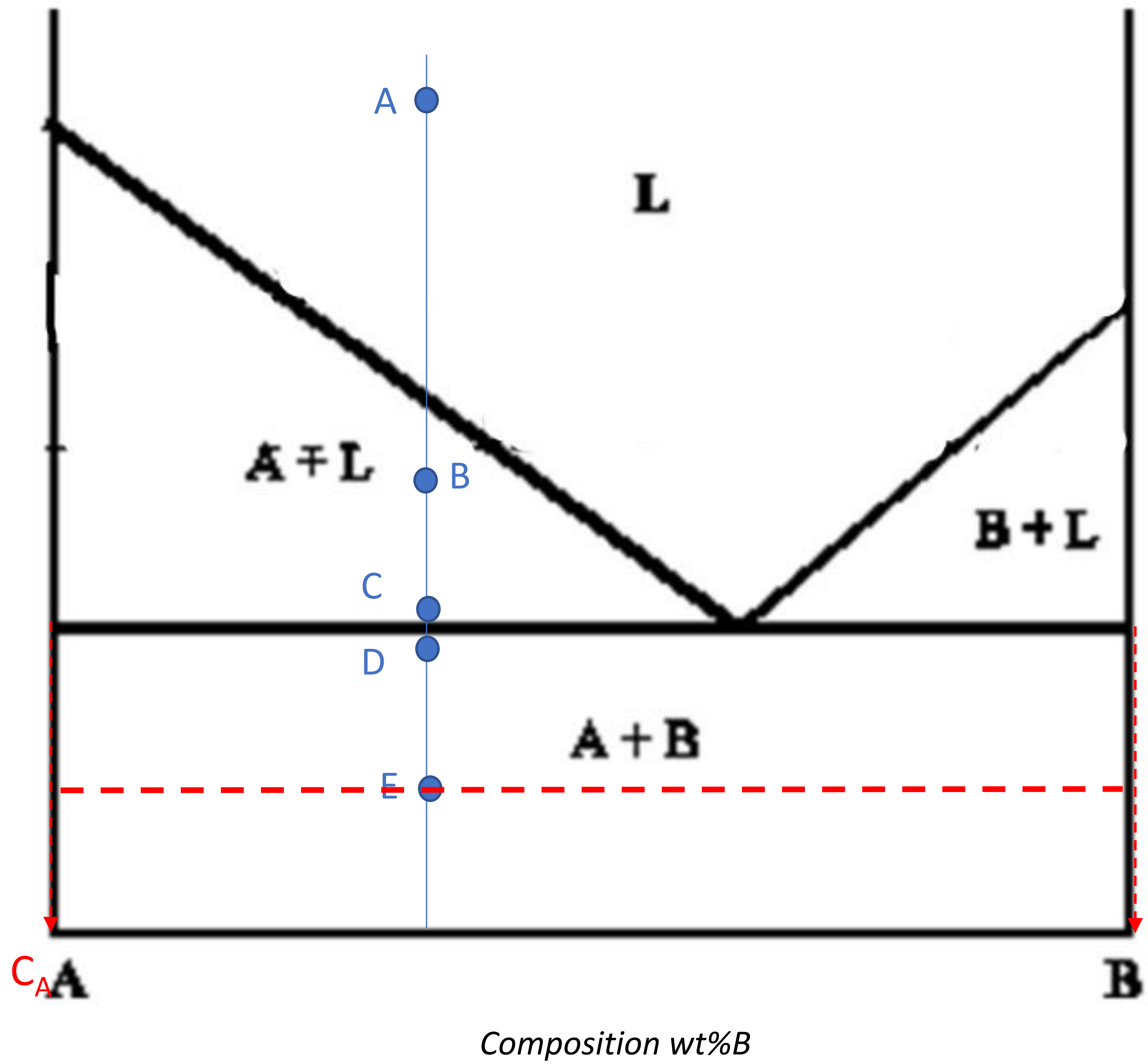
% A_{TOT} = (100-35)/(100-0)*100 = 65%

%B = (35-0)/(100-0)*100 = 35%



Primary/pro-eutectic A (As calculated in C, no more formed later): **% A**= (60-35)/(60-0)*100=42%

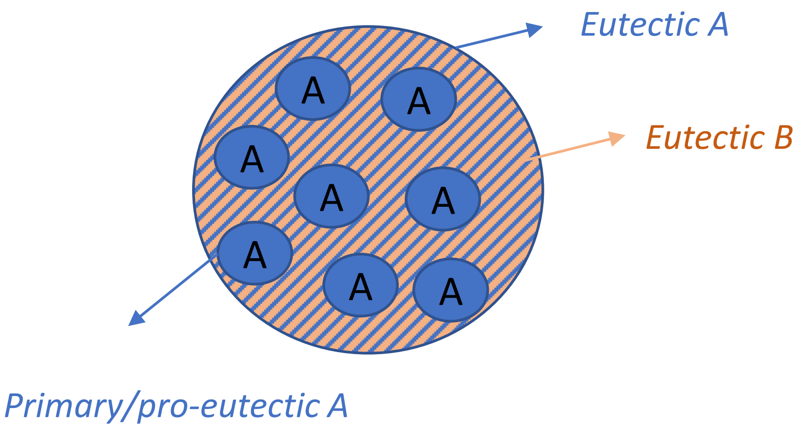
$\%A_{EUT}$= 65-42=23%



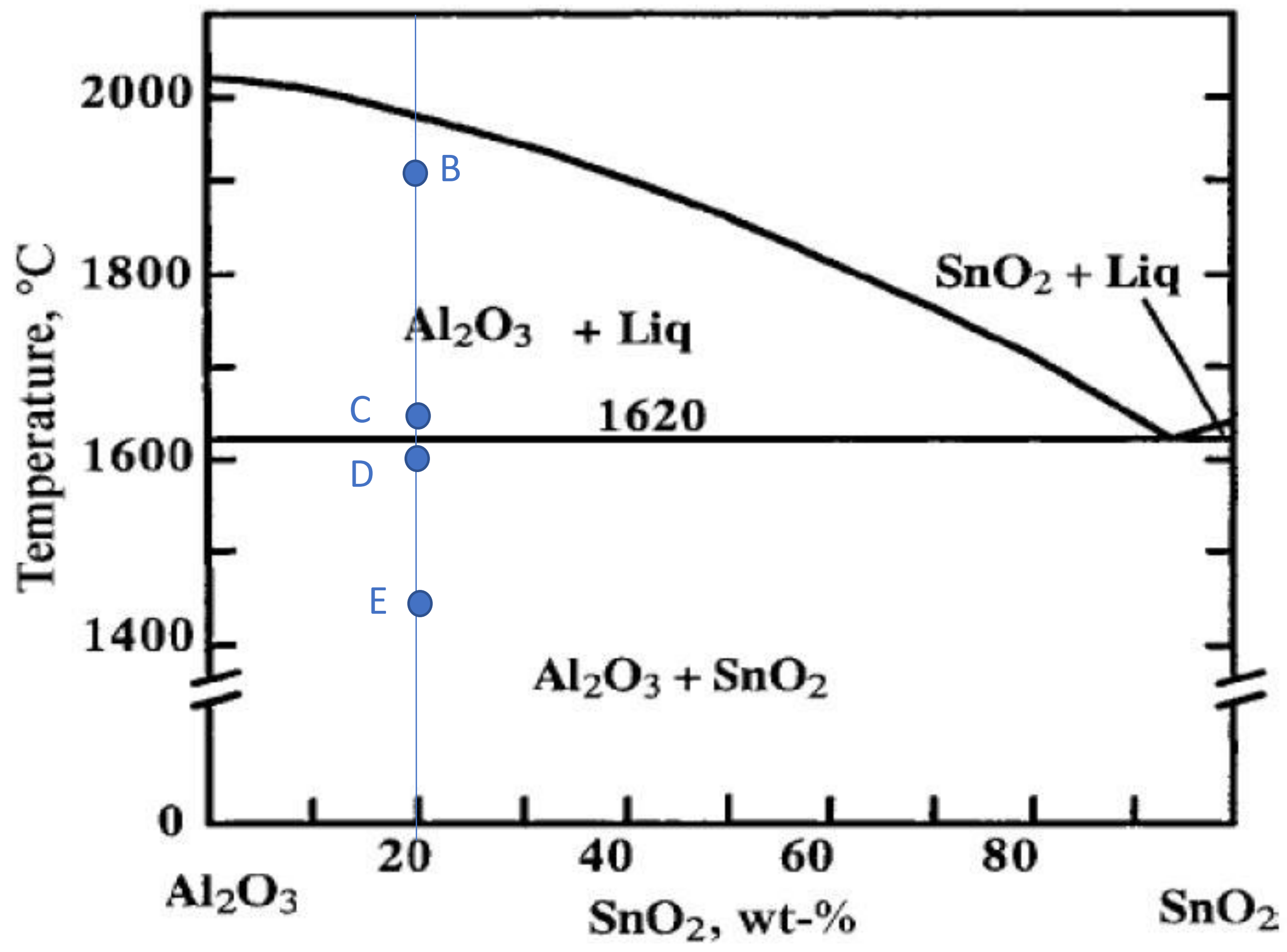
<u>Point E:</u>
Phases: 2 pahses solid (A) + solid (B)

Composition:
2 phases region → <u>horizontal rule</u>
Composition of Solid (A): C_A= 100%A
Composition of solid (B): C_B= 100%B

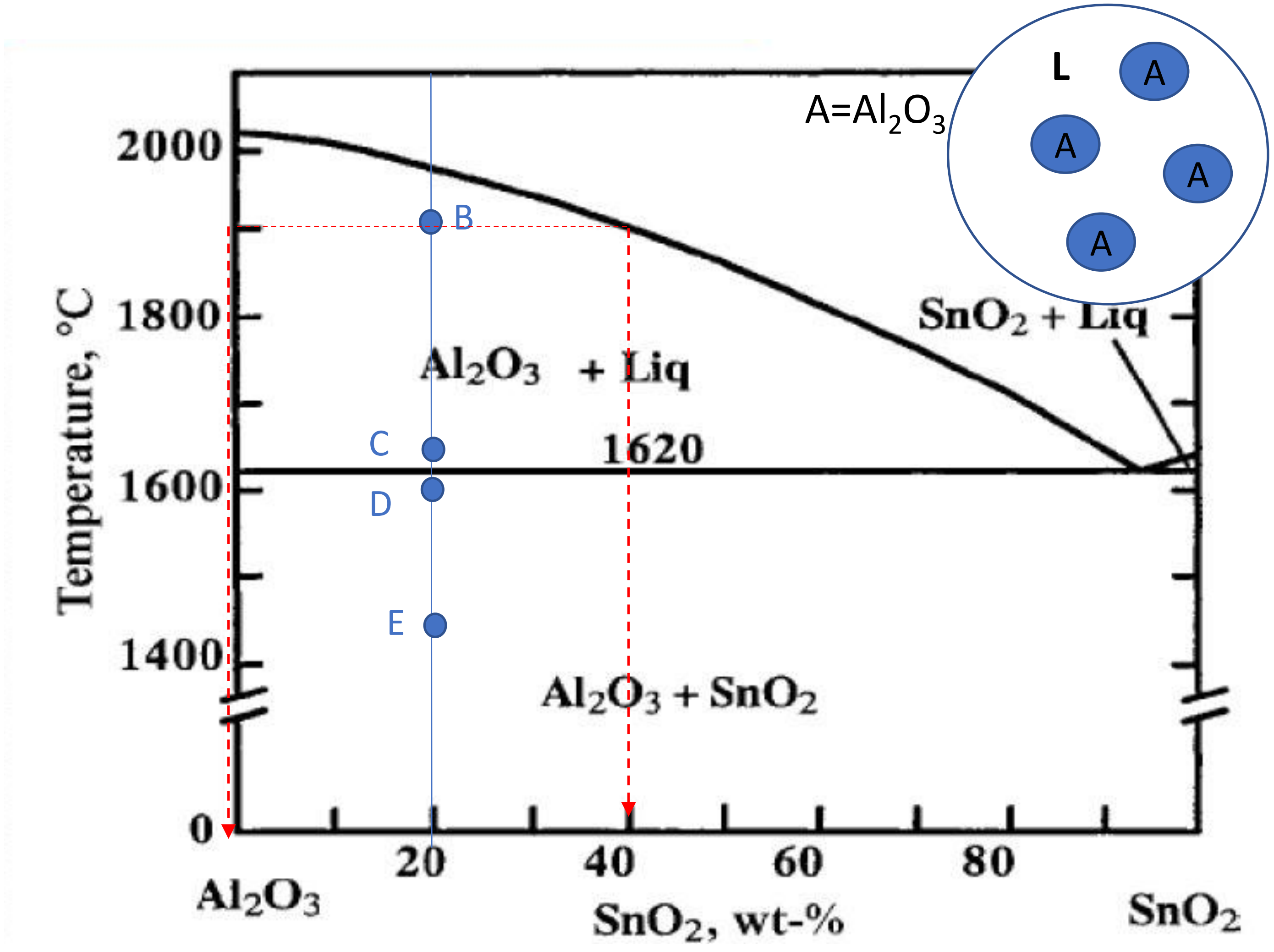
Relative amount:
2 phases region → <u>lever rule</u>
% A_{TOT} = (100-35)/(100-0)*100 = 65%
%B = (35-0)/(100-0)*100 = 35%



Exercise 2: determine the phase present, their composition and relative amounts in the points B, C (at the beginning of the eutectic transformation), D (at the end of the eutectic transformation) and E indicated in the phase diagram. Schematize the microstructure in each point.



Exercise 2: determine the phase present, their composition and relative amounts in the points B, C (at the beginning of the eutectic transformation), D (at the end of the eutectic transformation) and E indicated in the phase diagram. Schematize the microstructure in each point.



Point B:

Phases: 2 pahses solid (Al_2O_3) +liquid (L)

Composition:

2 phases region → horizontal rule

Composition of liquid (L): = 40% SnO_2, 60%Al_2O_3

Composition of solid: = 100% Al_2O_3

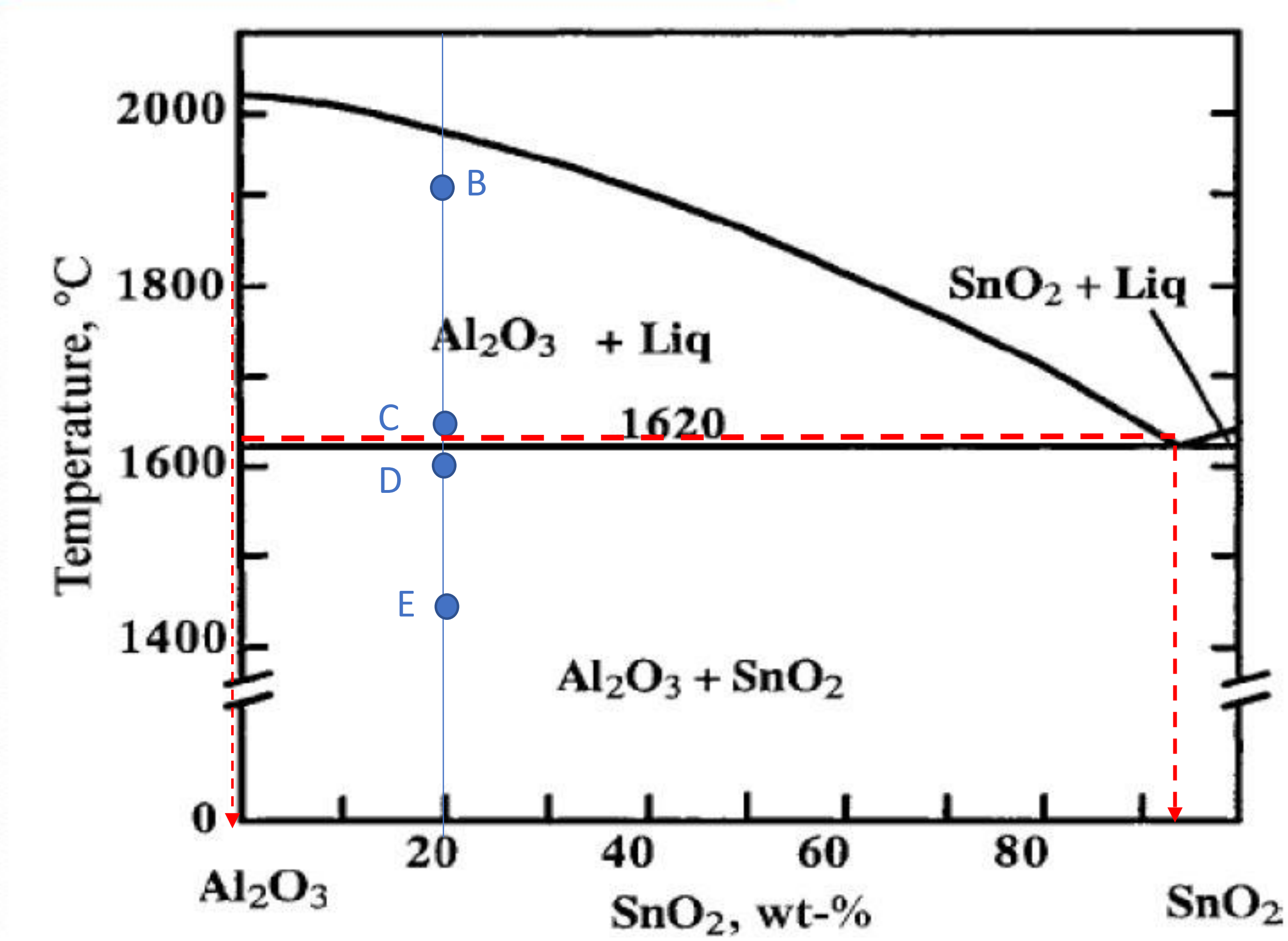
Relative amount:

2 phases region → lever rule

% Al_2O_3= (40-20)/(40-0)*100 = 50%

%L= (20-0)/(40-0)*100 = 50%

Exercise 2: determine the phase present, their composition and relative amounts in the points B, C (at the beginning of the eutectic transformation), D (at the end of the eutectic transformation) and E indicated in the phase diagram. Schematize the microstructure in each point.



Point C:

Phases: 2 pahses solid (Al_2O_3) +liquid (L)

Composition:

2 phases region → <u>horizontal rule</u>

Composition of liquid (L): = 92% SnO_2, 8%Al_2O_3

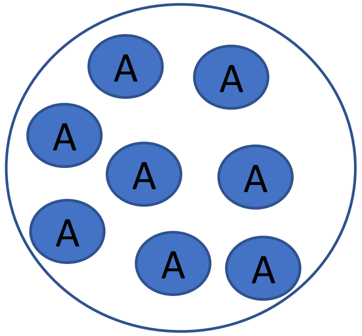
Composition of solid: = 100% Al_2O_3

Relative amount:

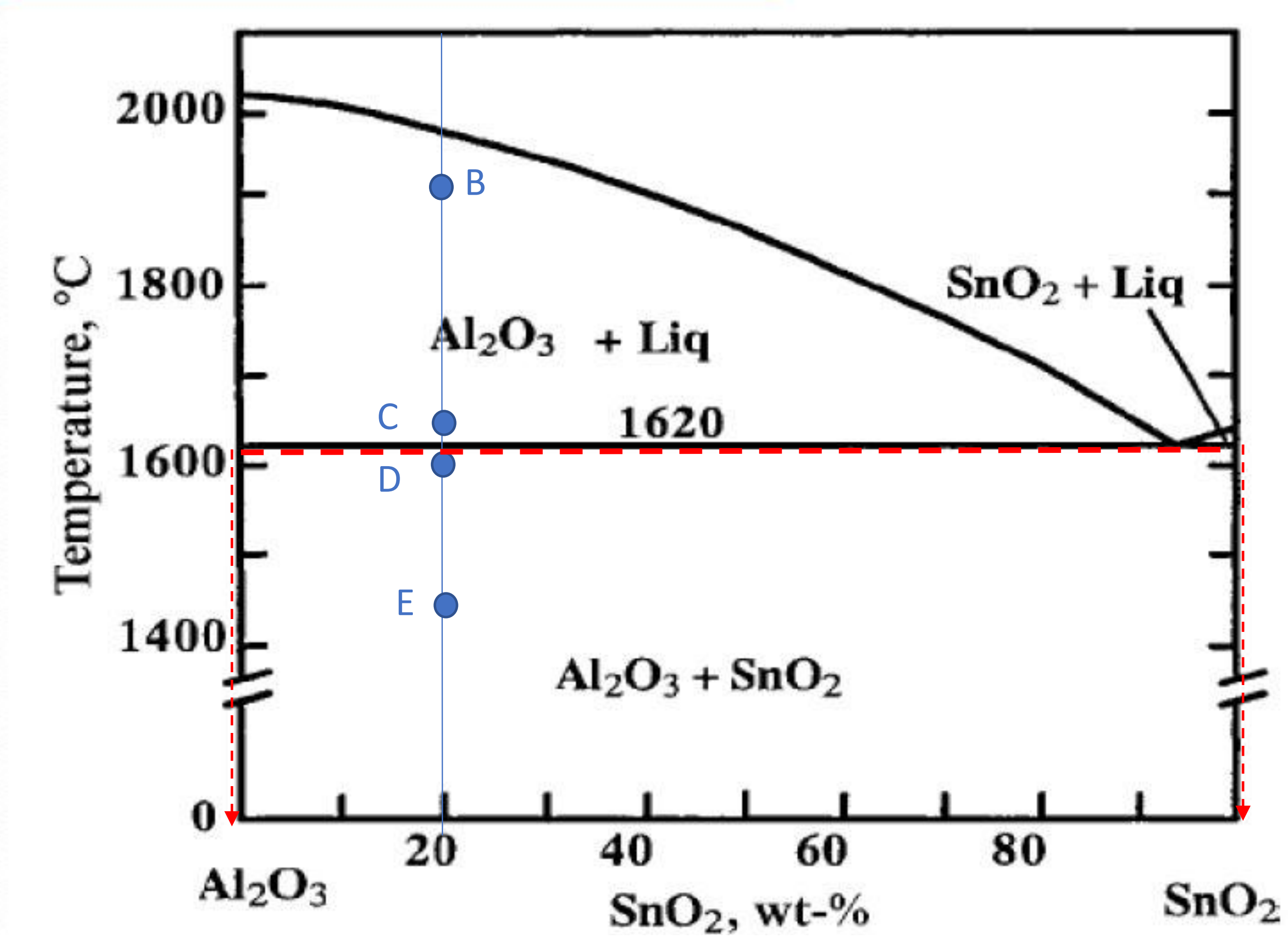
2 phases region → <u>lever rule</u>

% Al_2O_3= (92-20)/(92-0)*100 = 78%

%L = (20-0)/(92-0)*100 = 22%



A=Al_2O_3

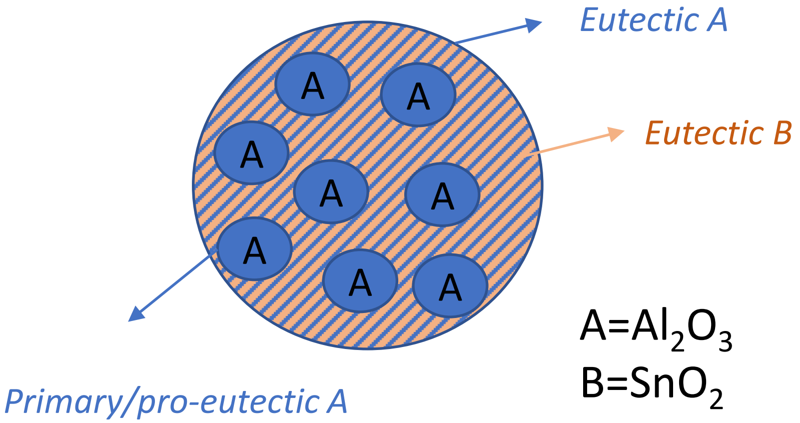


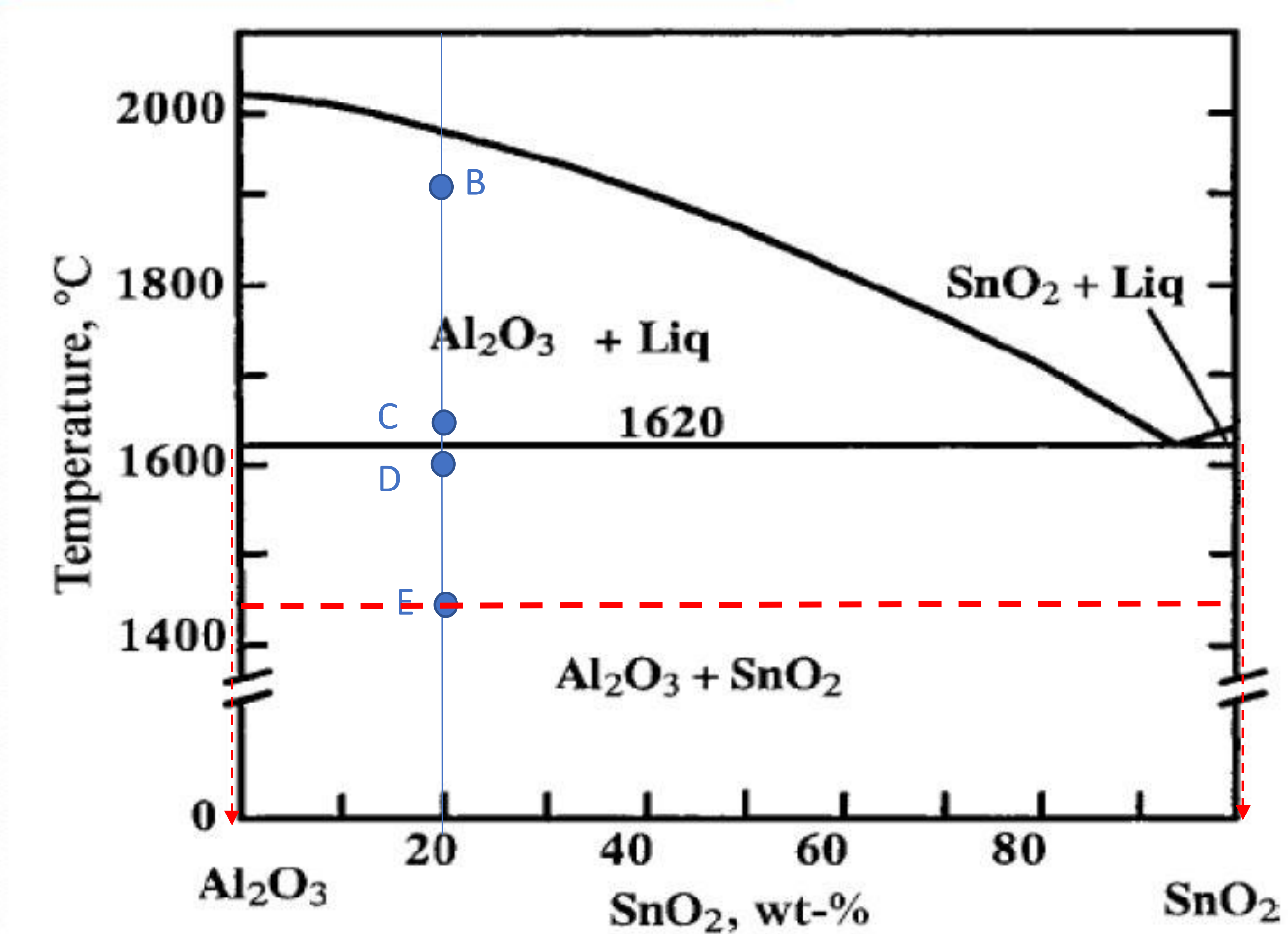
Primary/pro-eutectic Al_2O_3 (As calculated in C, no more formed later): **% Al_2O_3 $_{PRIM}$**= (92-20)/(92-0)*100=78%
$\%A_{EUT}$= 80-78=2%

Point D:
Phases: 2 pahses solid (Al_2O_3) + solid (SnO_2)

Composition:
2 phases region $\rightarrow$ horizontal rule
Composition of Solid A = 100% Al_2O_3
Composition of solid (B): C_B= 100% SnO_2

Relative amount:
2 phases region $\rightarrow$ lever rule
% $Al_2O_{3\ TOT}$ = (100-20)/(100-0)*100 = 80%
%SnO_2 = (20-0)/(100-0)*100 = 20%





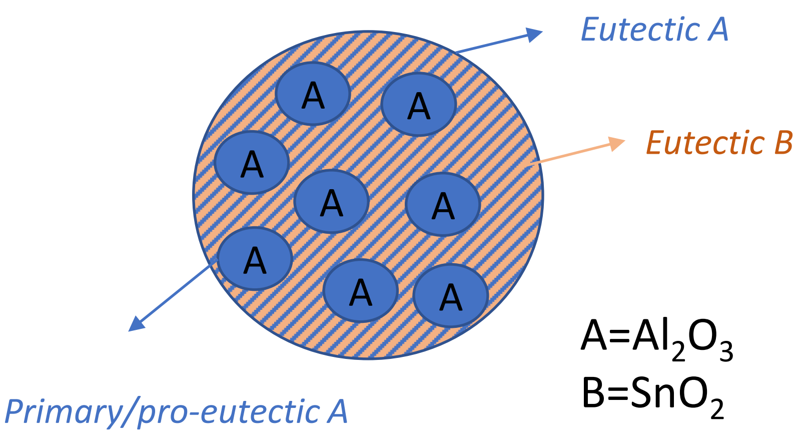
<u>Point E:</u>
Phases: 2 pahses solid (Al_2O_3) + solid (SnO_2)

Composition:
2 phases region → <u>horizontal rule</u>

Composition of Solid A = 100%Al_2O_3
Composition of solid (B): C_B= 100%SnO_2

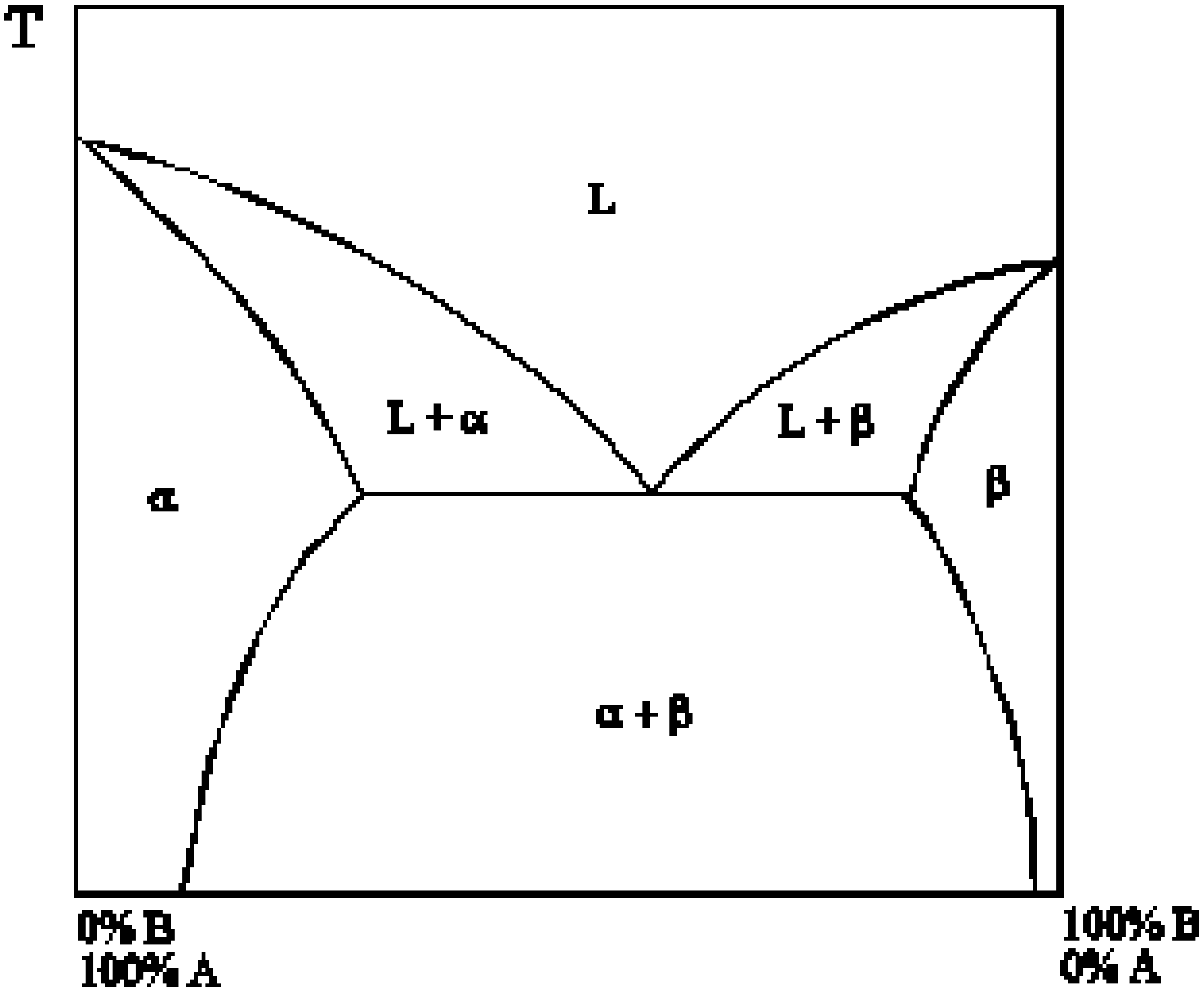
Relative amount:
2 phases region → <u>lever rule</u>

% Al_2O_3 $_{TOT}$ = (100-20)/(100-0)*100 = 80%
%SnO_2= (20-0)/(100-0)*100 = 20%

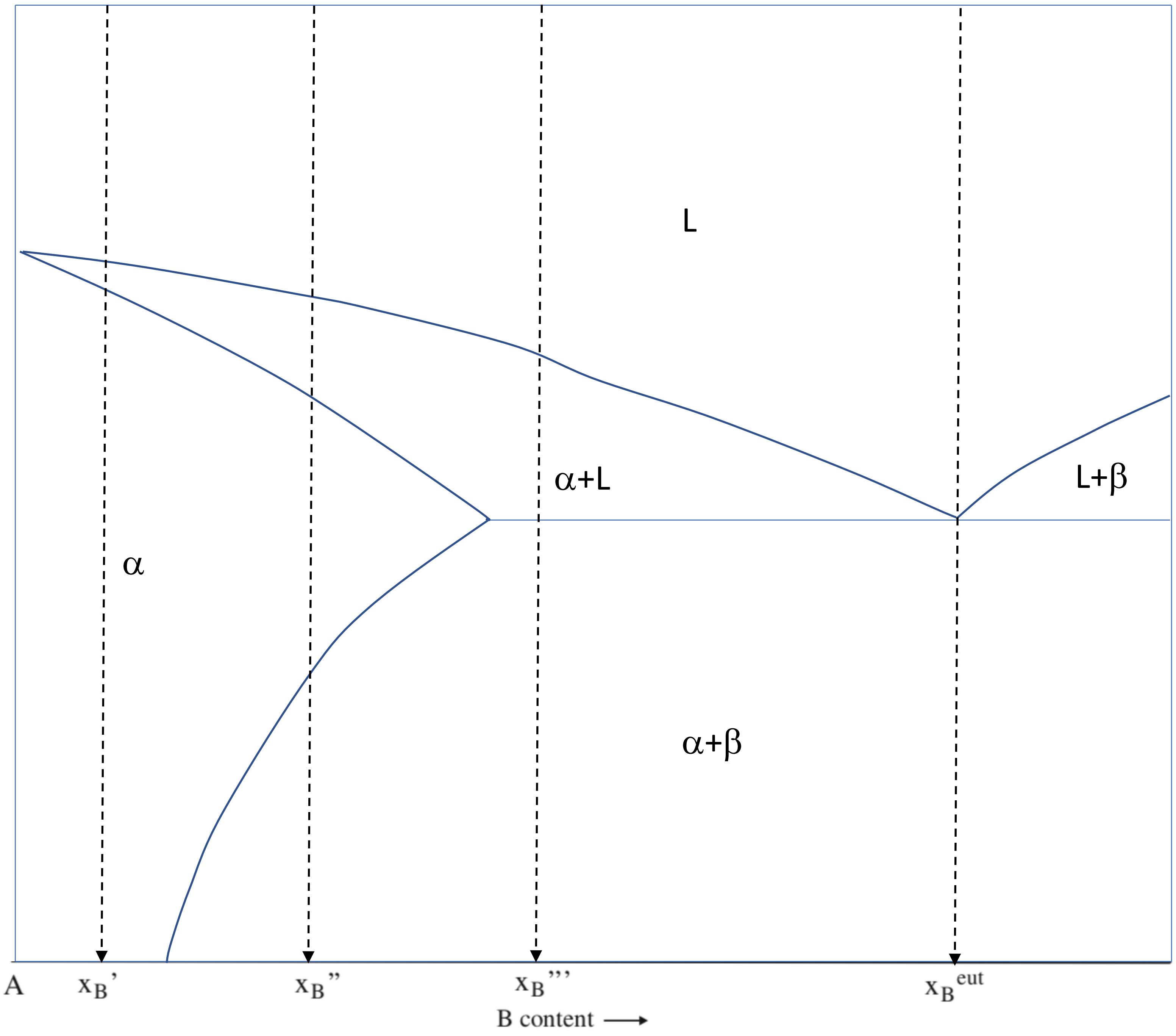


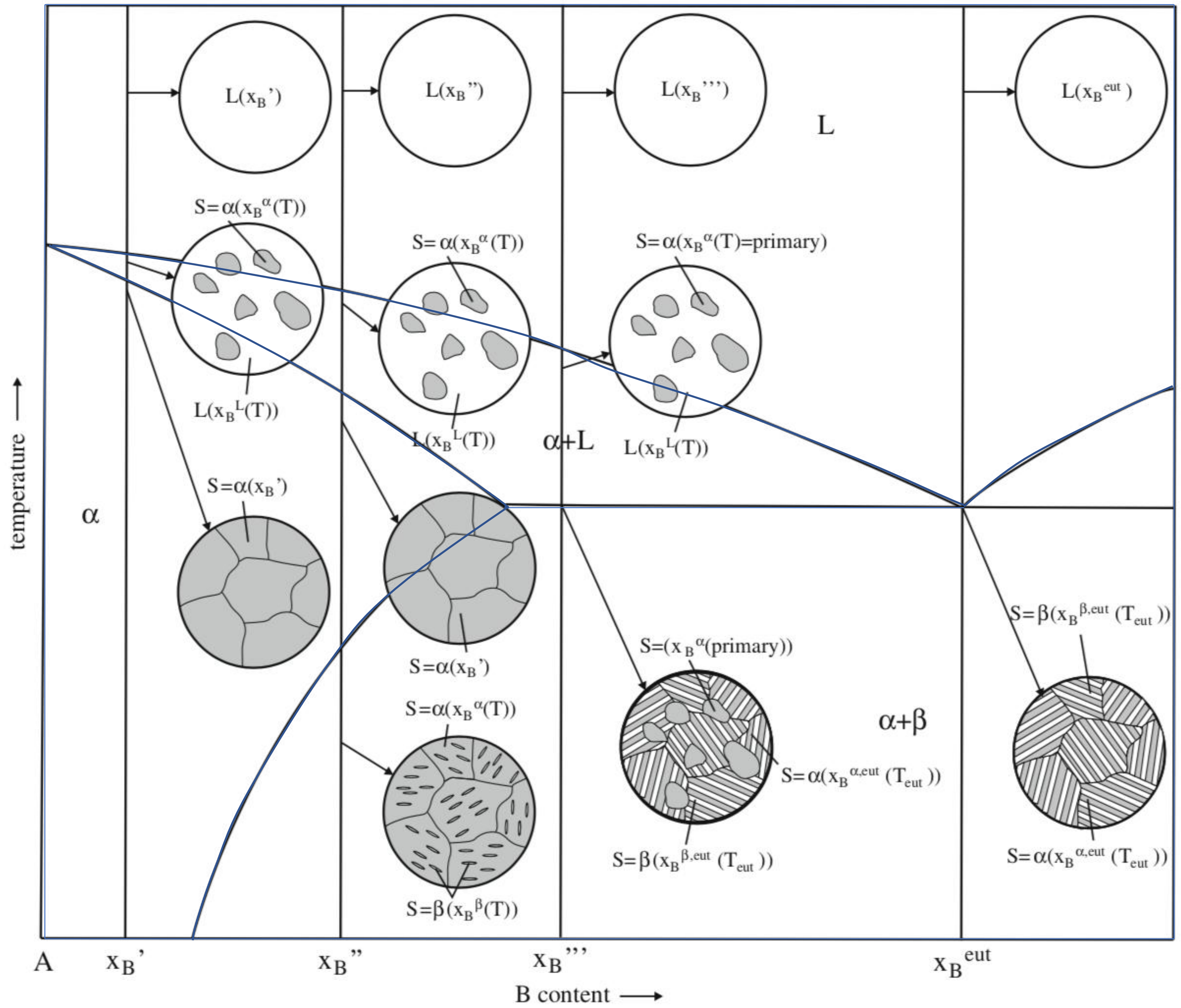
Exercise 3: Draw a phase diagram (for generic components A and B), with complete solubility in the liquid state and **partial solubility in the solid state**. Also indicate the phase/phases present(s) in each region of the phase diagram

Complete solubility in the liquid state, partial solubility in the solid state



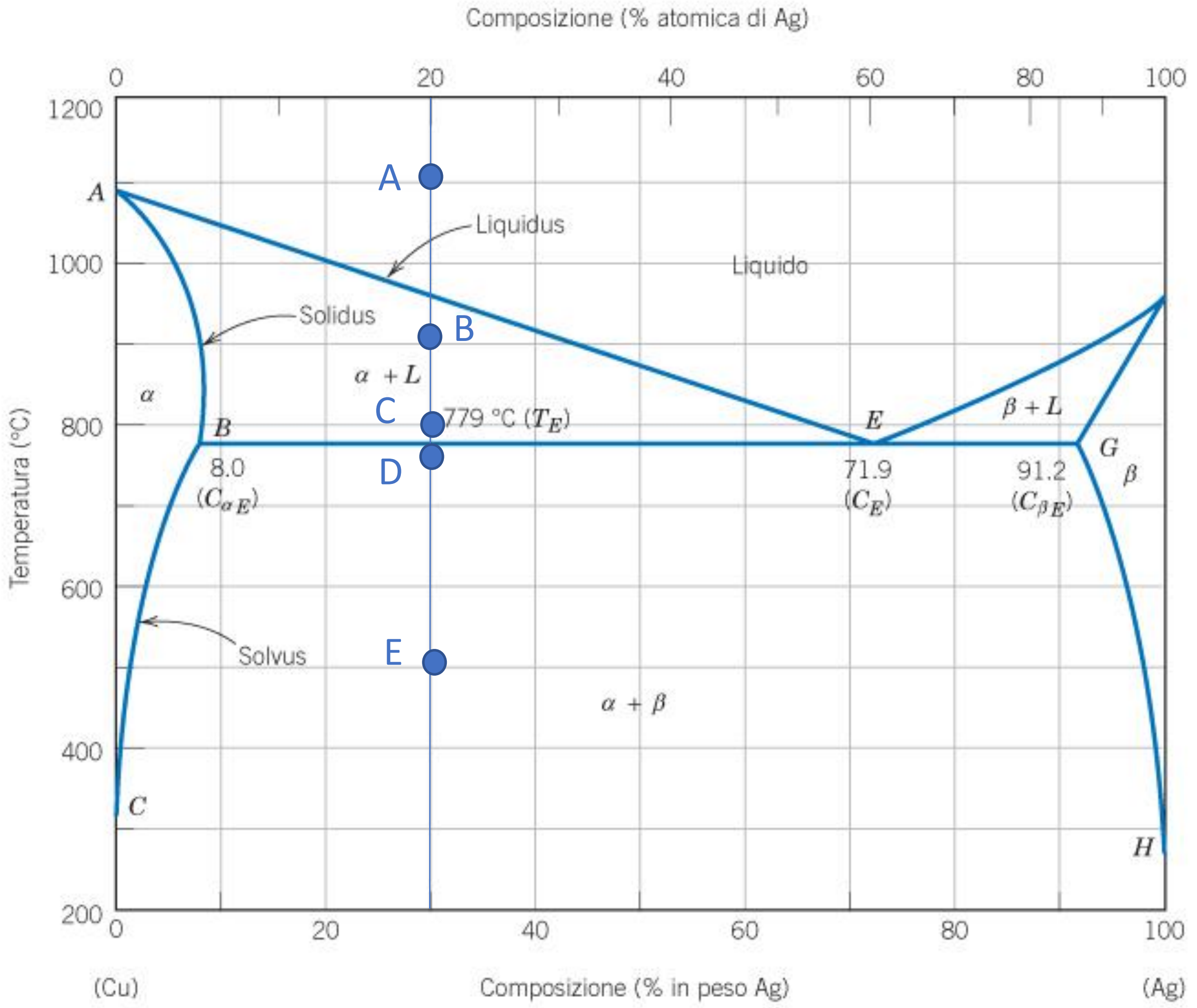
Exercise 4: describe the solidification of alloys with compositions x_B', x_B'', x_B''' ans x_B^{eut} by drawing schematic microstructures in significant points.

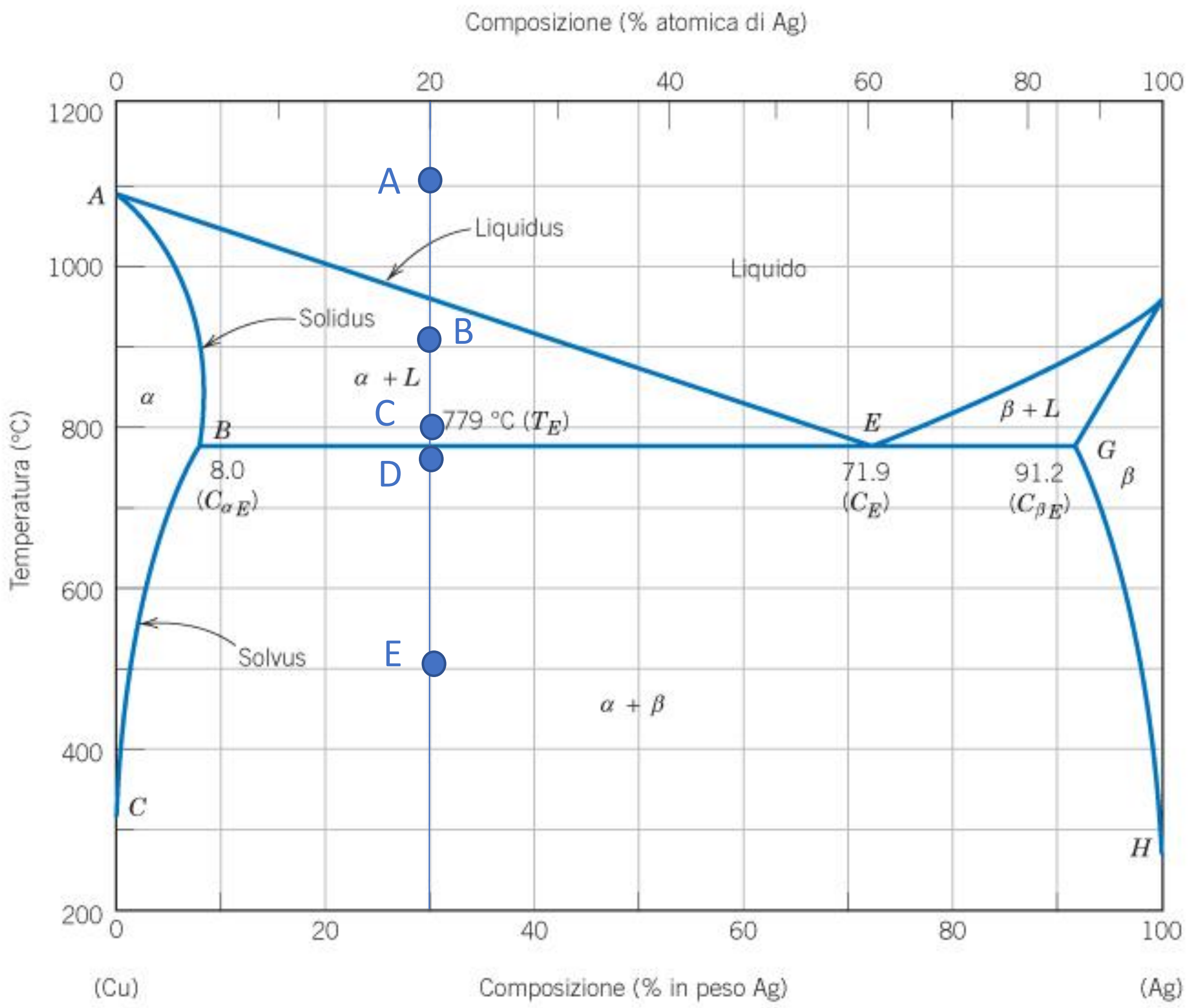




L(x_B')
L(x_B'')
L(x_B''')
L
L(x_B^eut)
S=α(x_B^α(T))
S=α(x_B^α(T))
S=α(x_B^α(T)=primary)
L(x_B^L(T))
L(x_B^L(T))
α+L
L(x_B^L(T))
S=α(x_B')
α
S=α(x_B')
S=α(x_B^α(T))
S=β(x_B^β(T))
S=(x_B^α(primary))
α+β
S=α(x_B^α,eut (T_eut))
S=β(x_B^β,eut (T_eut))
S=β(x_B^β,eut (T_eut))
S=α(x_B^α,eut (T_eut))
temperature
A
x_B'
x_B''
x_B'''
x_B^eut
B content

Exercise 5: determine the phase present, their composition and relative amounts in the points A, B, C (at the beginning of the eutectic transformation), D (at the end of the eutectic transformation) and E indicated in the phase diagram. Schematize the microstructure in each point.



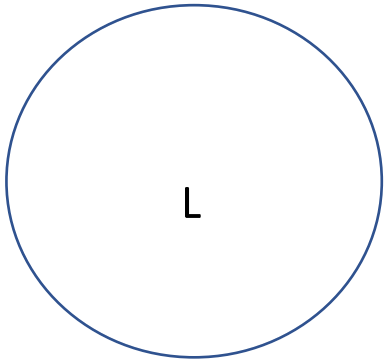


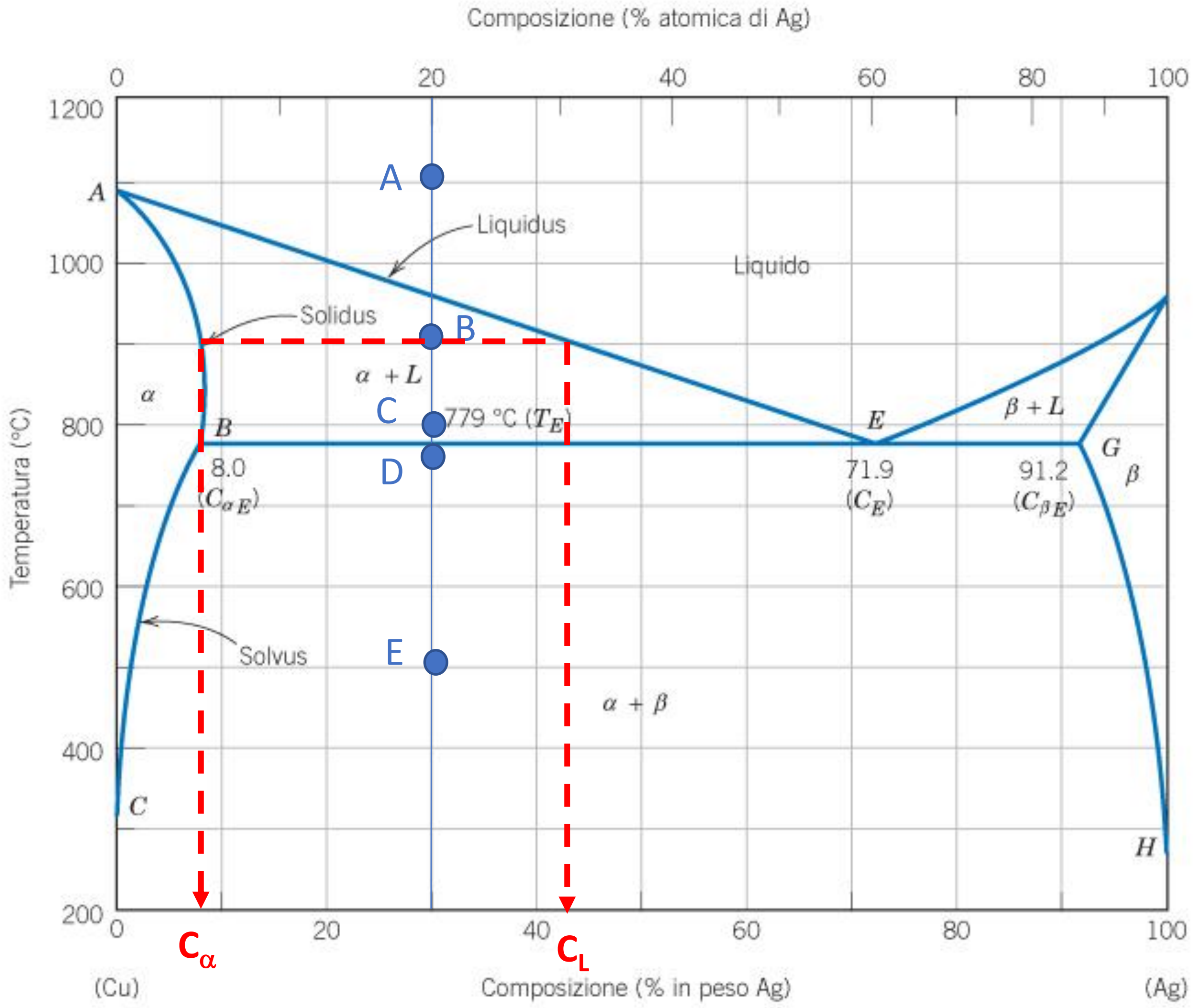
<u>Point A:</u>

Phases: 1 phase liquid

Composition: 30%Ag, 70%Cu

Relative amount: 100% liquid phase





Point B:

Phases: 2 pahses solid (α) +liquid (L)

Composition:

2 phases region → horizontal rule

Composition of liquid (L): C_L= 42% Ag, 58% Cu

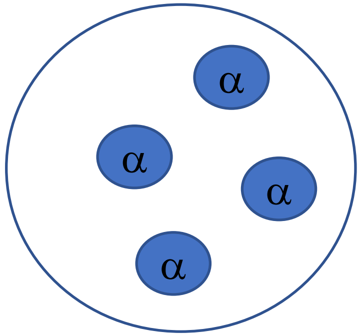
Composition of solid (α): C_α= 8% Ag, 92% Cu

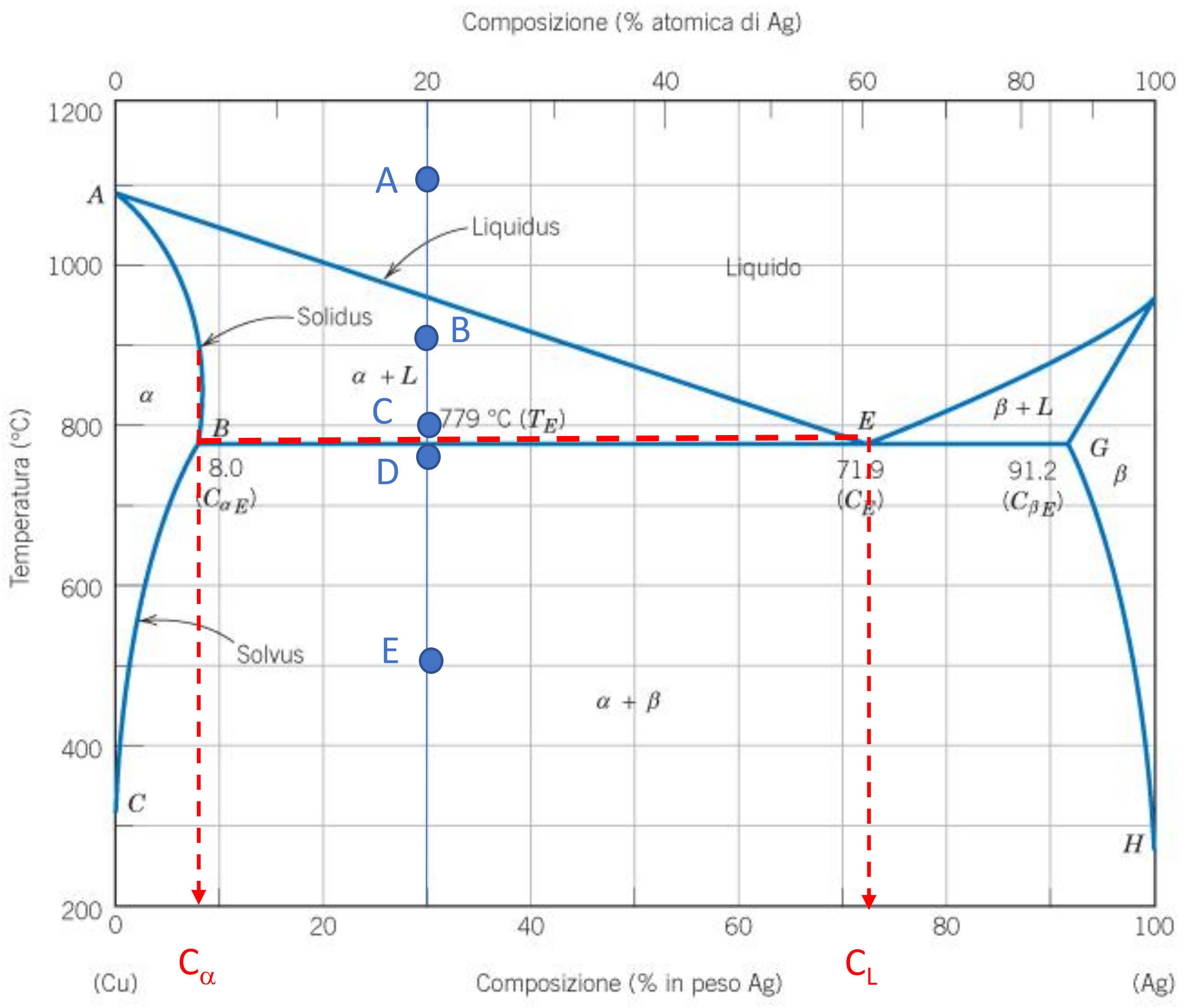
Relative amount:

2 phases region → lever rule

% α = (42-30)/(42-8)*100 = 35%

%L = (30-8)/(42-8)*100 = 65%





<u>Point C:</u>

Phases: 2 phases solid (α) +liquid (L)

Composition:

2 phases region → <u>horizontal rule</u>

Composition of liquid (L): C_L= 71.9% Ag, 28.1% Cu

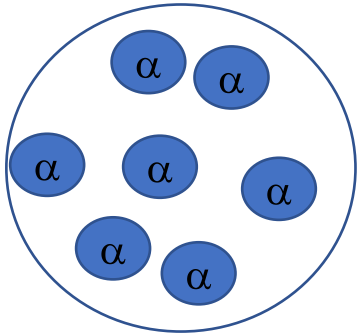
Composition of solid (α): C_α= 8% Ag, 92% Cu

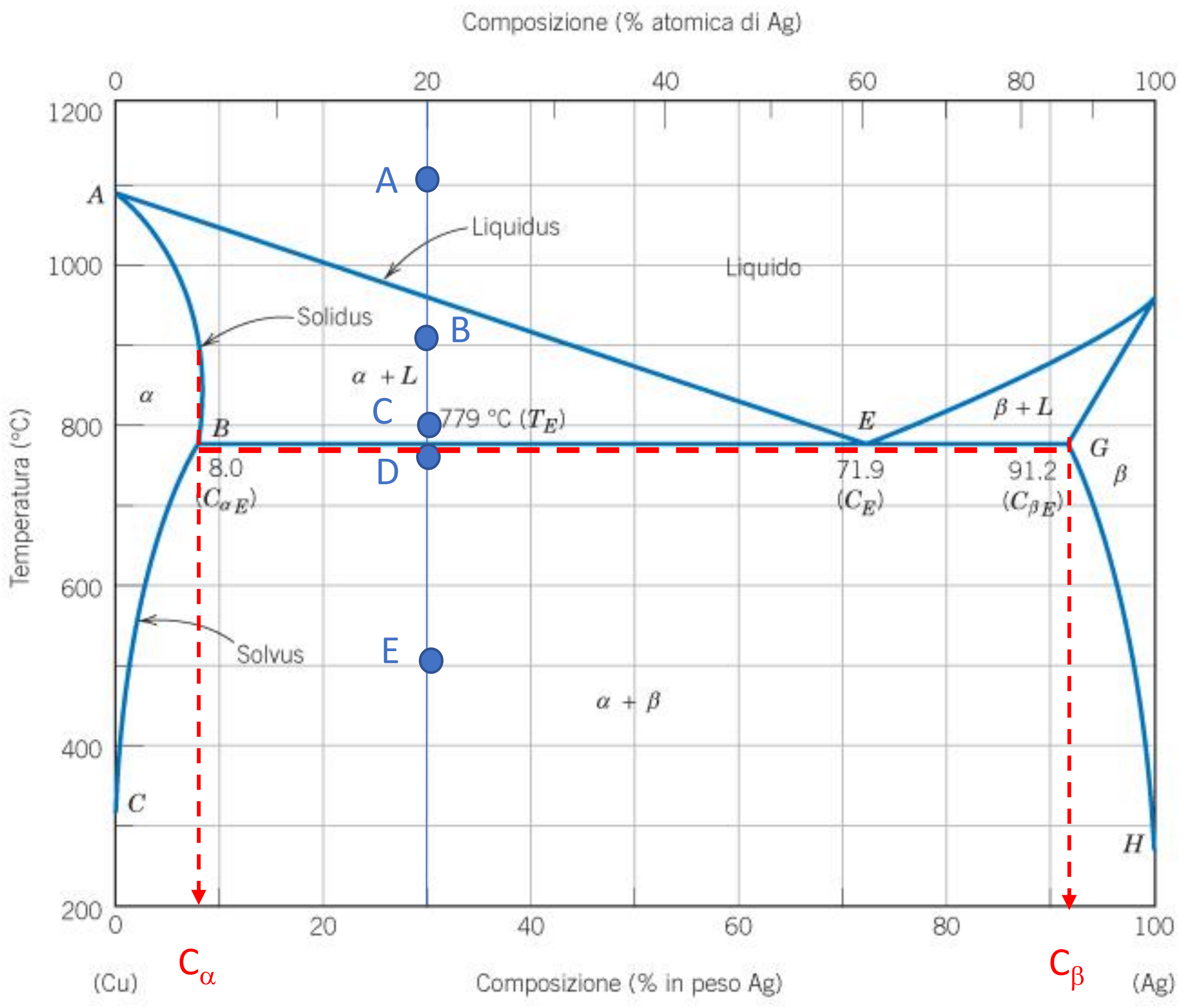
Relative amount:

2 phases region → <u>lever rule</u>

% α = (71.9-30)/(71.9-8)*100 = 65.6%

%L = (30-8)/(71.9-8)*100 = 34.4%





Point D:

Phases: 2 pahses solid (α) +solid (β)

Composition:

2 phases region → horizontal rule

Composition of solid (α): C_α= 8% Ag, 92% Cu

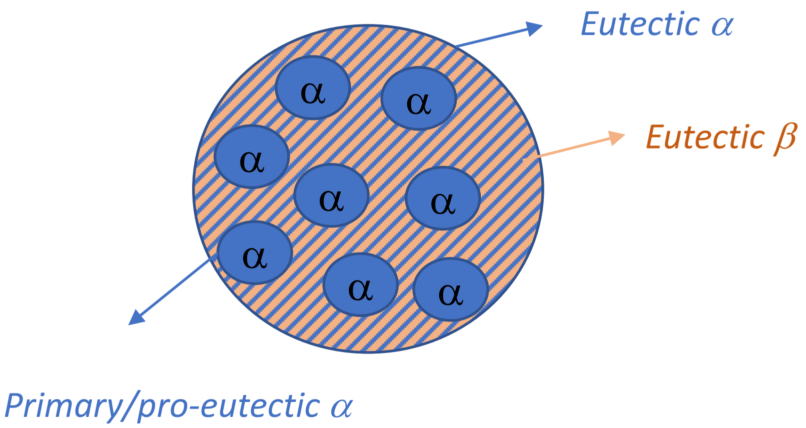
Composition of solid (β): Cβ= 91.2% Ag, 8.8% Cu

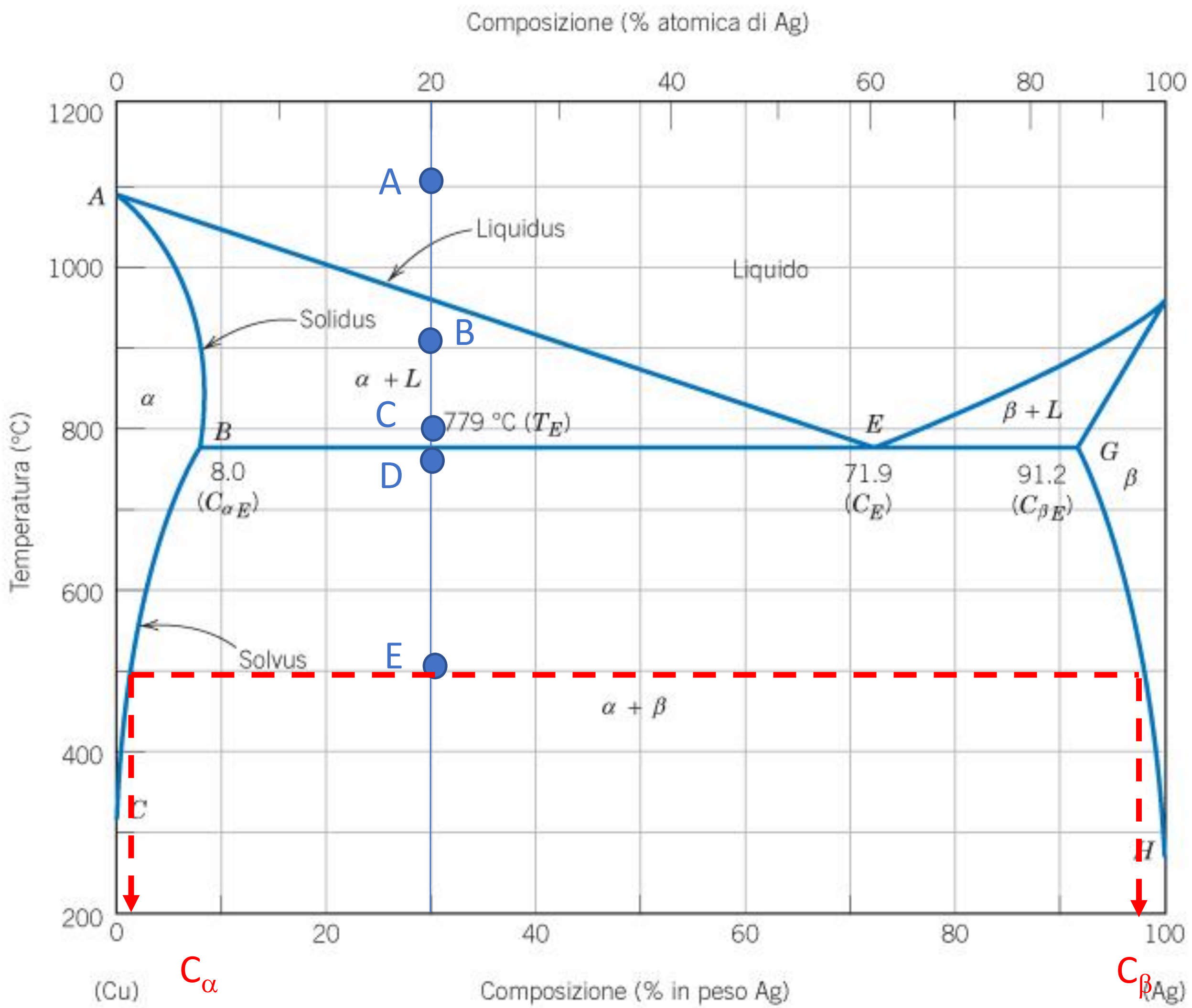
Relative amount:

2 phases region → lever rule

% α = (91.2-30)/(91.2-8)*100 = 73.5%

%β = (30-8)/(91.2-8)*100 = 26.5%





Point E:

Phases: 2 pahses solid (α) +solid (β)

Composition:

2 phases region → horizontal rule

Composition of solid (α): C_α= 2% Ag, 98% Cu

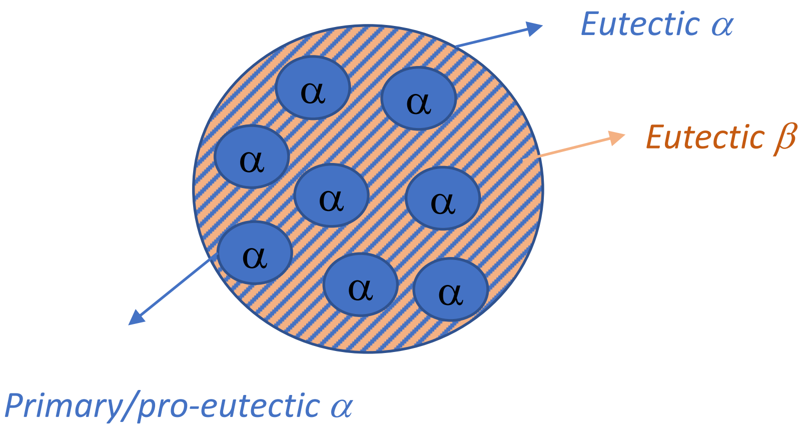
Composition of solid (β): Cβ= 98% Ag, 2% Cu

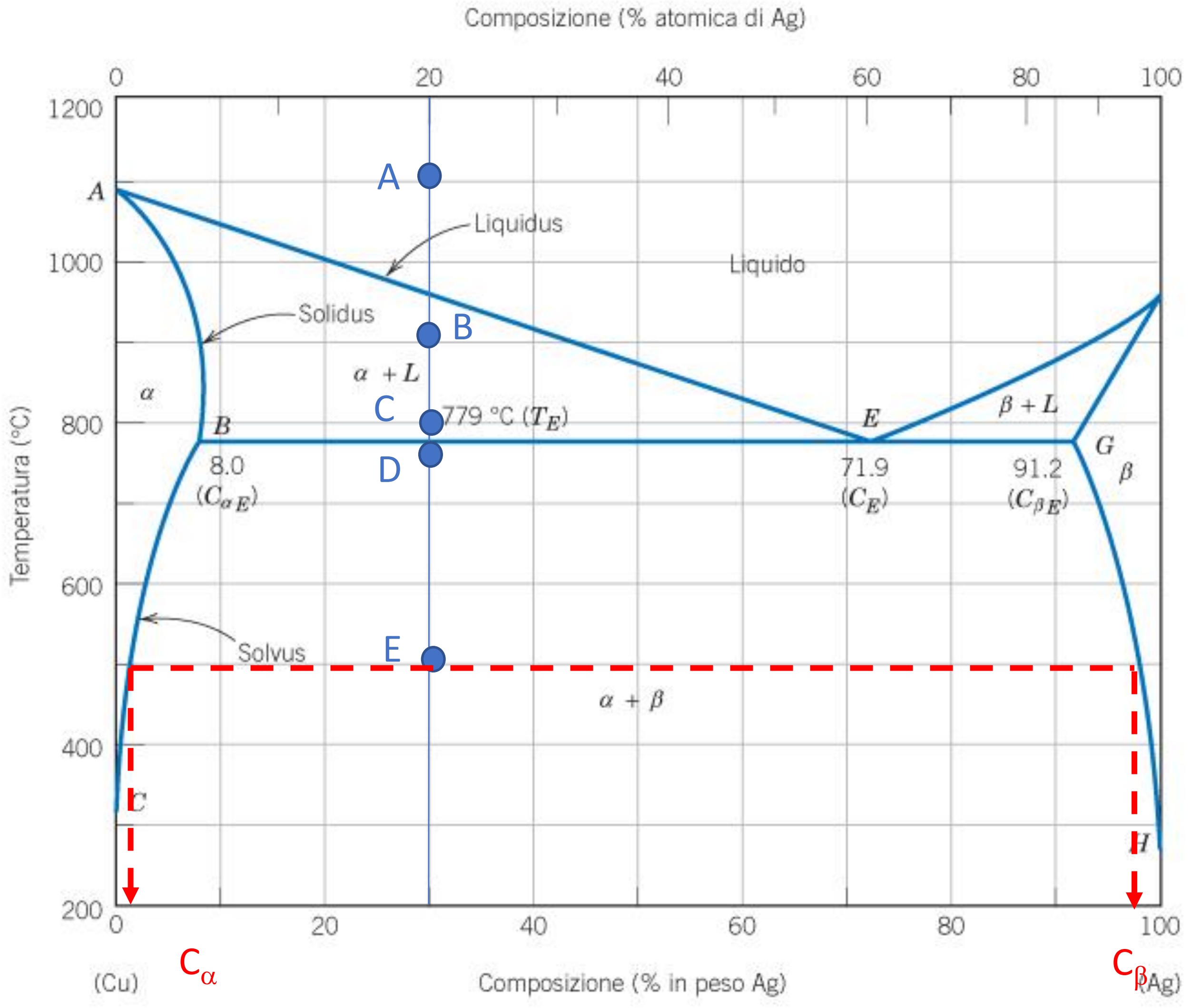
Relative amount:

2 phases region → lever rule

% α = (98-30)/(98-2)*100 = 71%

%β = (30-2)/(98-2)*100 = 29%





<u>Point E:</u>

Phases: 2 pahses solid (α) +solid (β)

Composition:

2 phases region → <u>horizontal rule</u>

Composition of solid (α): C_α= 2% Ag, 98% Cu

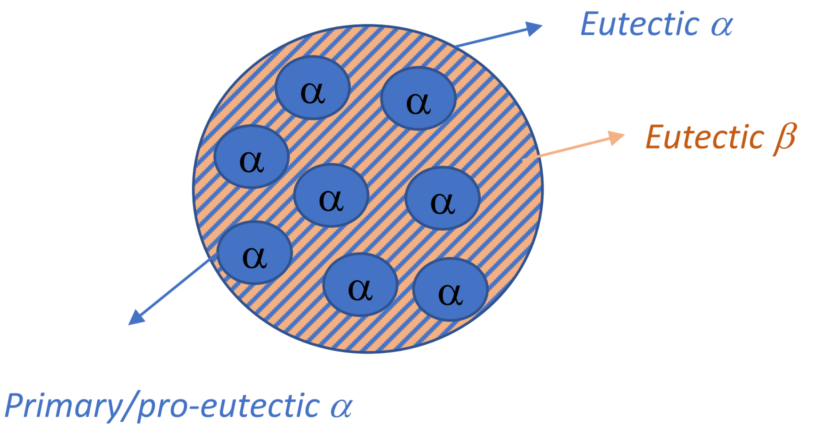
Composition of solid (β): Cβ= 98% Ag, 2% Cu

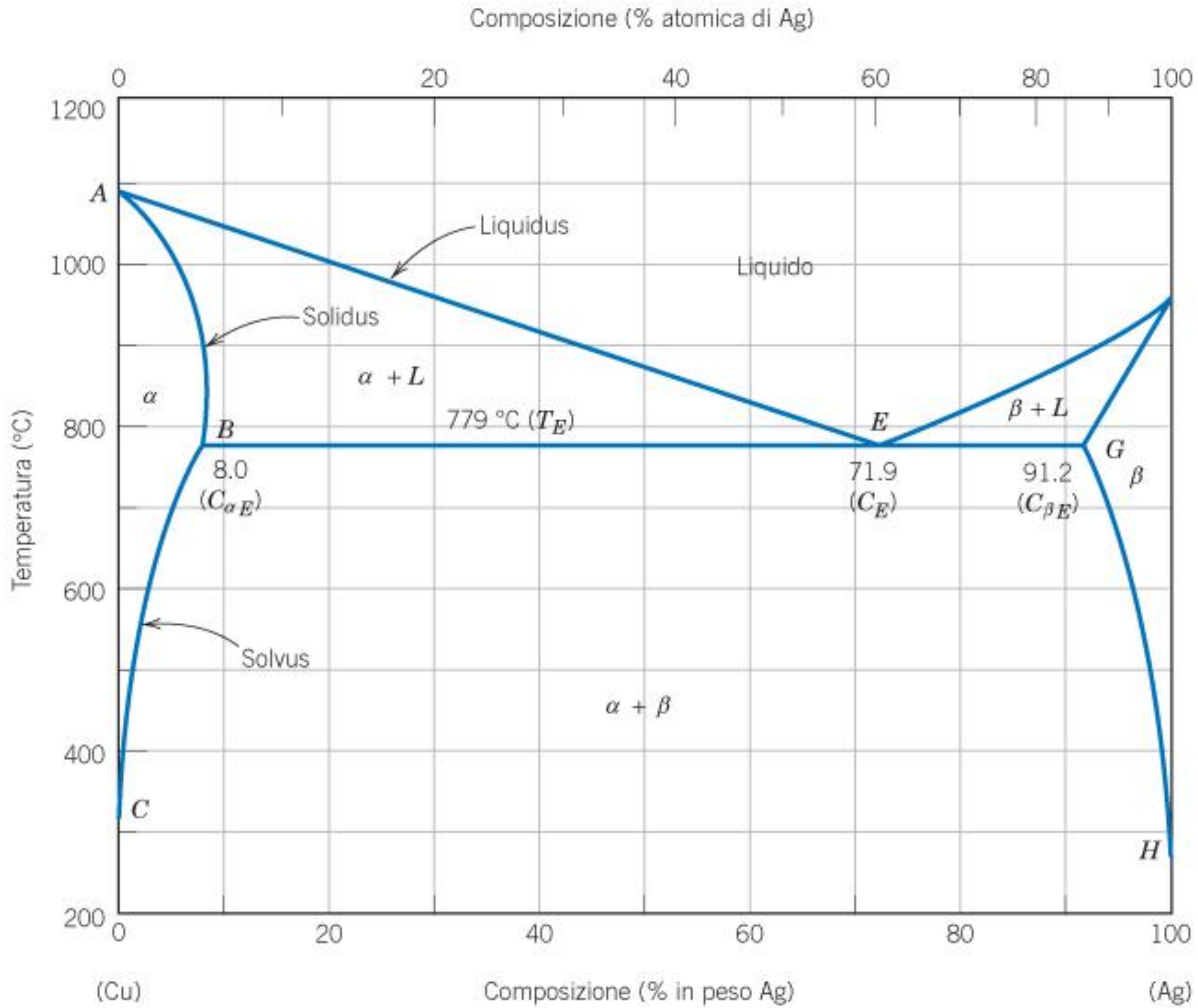
Relative amount:

2 phases region → <u>lever rule</u>

% α = (98-30)/(98-2)*100 = 71%

%β = (30-2)/(98-2)*100 = 29%




Composizione (% atomica di Ag)
0
20
40
60
80
100
1200
1000
800
600
400
200
Temperatura (°C)
A
Liquidus
Liquido
Solidus
α
α + L
B
779 °C (T_E)
E
β + L
G
β
8.0
($C_{\alpha E}$)
71.9
(C_E)
91.2
($C_{\beta E}$)
Solvus
α + β
C
H
0
20
40
60
80
100
(Cu)
Composizione (% in peso Ag)
(Ag)

Eutectic phase diagram